전국기능경기대회 채점기준

1. 채점 시 유의사항

직 종 명	클라우드컴퓨팅

※ 다음 사항을 유의하여 채점하시오.

1) AWS의 지역은 ap-northeast-2을 사용합니다.
2) 웹페이지 접근은 크롬이나 파이어폭스를 이용합니다.
3) 웹페이지에서 언어에 따라 문구가 다르게 보일 수 있습니다.
4) shell에서의 명령어의 출력은 버전에 따라 조금 다를 수 있습니다.
5) 채점은 문항 순서대로 진행해야 합니다.
6) 부분 점수가 따로 없는 문항은 모두 맞아야 점수로 인정됩니다.
7) 리소스의 정보를 읽어오는 채점항목은 기본적으로 스크립트 결과를 통해 채점을 진행하며, 만약 선수가 이의가 있다면 명령어를 직접 입력하여 확인해볼 수 있습니다.
8) (예상 출력)은 바로 이전 (명령어 입력)의 예상 출력을 의미합니다.
9) 채점 시에는 별도로 제공한 채점 스크립트(mark.sh)를 실행하여 채점할 수 있습니다.
다만, 선수가 직접 입력을 원할 경우 채점기준표에 명시된 명령어 그대로 입력하여 채점할 수 있습니다.
10) 제출 전 mark-sg의 경우 선수 임의로 구성해두도록 합니다.
11) 모든 채점 사항은 Cloudshell에 wsc2026-skills-app-sub-a 서브넷과 사전구성한 mark-sg를 연결한 후 진행합니다.
12) 11-1 채점을 실행하여 생긴 파드들은 채점이 끝난 후 삭제하도록 합니다.
13) 아래 명령어를 사전입력합니다.

```
aws configure set default.region ap-northeast-2
ACCOUNT_ID=$(aws sts get-caller-identity --query "Account" --output text)
VPC_ID=$(aws ec2 describe-vpcs --filters "Name=tag:Name,Values=wsc2026-skills-vpc" --query "Vpcs[0].VpcId" --output text)
BUCKET=$(aws s3api list-buckets --query "Buckets[?contains(Name,'wsc2026-static')].Name" --output text)
aws eks update-kubeconfig --name wsc2026-eks-cluster --region ap-northeast-2 2>/dev/null
check_kms() {
  local ALIAS=$1 ACTUAL_ARN=$2; local EXPECTED_ARN=$(aws kms describe-key --key-id "alias/${ALIAS}" --query "KeyMetadata.Arn" --output text 2>/dev/null); local KEY_ID=$(aws kms describe-key --key-id "alias/${ALIAS}" --query "KeyMetadata.KeyId" --output text 2>/dev/null)
  if [ -z "$KEY_ID" ] || [ "$KEY_ID" = "None" ]; then echo "KMS ${ALIAS}: FAIL (key not found)"; return; fi
  if [ "$EXPECTED_ARN" != "$ACTUAL_ARN" ]; then echo "KMS ${ALIAS}: FAIL (wrong key)"; return; fi
  POLICY=$(aws kms get-key-policy --key-id "$KEY_ID" --policy-name default --output text 2>/dev/null)
  if echo "$POLICY" | grep -q '"kms:W*"'; then echo "KMS ${ALIAS}: FAIL (kms:*)"; elif echo "$POLICY" | grep -q ':root"'; then echo "KMS ${ALIAS}: FAIL (root)"; else echo "KMS ${ALIAS}: PASS"; fi
}
```

※ 채점기준 양식은 반드시, 양식에 맞추어 작성해야 합니다.(채점사이트 입력에 필요)

2. 채점기준표

1) 주요항목별 배점

직 종 명	클라우드컴퓨팅

과제 번호	일련 번호	주요항목	배점	채점방법		채점시기		비고
				독립	합의	경기 진행중	경기 종료후	
제1과제	1	Networking	2.0	○			○	
	2	Database	1.3	○			○	
	3	Container Registry	1.2	○			○	
	4	Container Orchestration	3.0	○			○	
	5	Deployment	6.5	○			○	
	6	S3	1.0	○			○	
	7	Lambda	3.0	○			○	
	8	ALB	1.3	○			○	
	9	CloudFront	4.0	○			○	
	10	WAF	1.5	○			○	
	11	Observability	5.2	○			○	
합 계			30					

2) 채점방법 및 기준

과제 번호	일련 번호	주요항목	일련 번호	세부항목(채점방법)	배점
제1과제	1	Networking	1	VPC	1.0
			2	Routing	1.0
	2	Database	1	DynamoDB Table	1.3
	3	Container Registry	1	ECR Container	1.2
	4	Container Orchestration	1	EKS Cluster	1.5
			2	NodeGroup	1.0
			3	EKS IAM	0.5
	5	Deployment	1	Resource Status	1.5
			2	Scheduling & Resources	1.0
			3	Probes & ConfigMap	1.0
			4	Service & Pod Placement	1.5
			5	Pod Identity	1.5
	6	S3	1	S3 Bucket	1.0
	7	Lambda	1	Lambda Functions	1.5
			2	Lambda IAM	1.5
	8	Load Balancer	1	Load Balancer	1.3
	9	CloudFront	1	CloudFront Distribution	1.2
			2	Cache Policy	1.3
			3	POST & GET E2E	1.5
	10	WAF	1	WAF Protection	1.5
	11	Observability	1	Observability Deploy	1.2
			2	Grafana Datasource	1.0
			3	Grafana Dashboard	1.5
			4	Alert Firing	1.5
합계					30

순번	채점 항목
1-1	1) aws ec2 describe-vpcs --vpc-ids "$VPC_ID" --query "Vpcs[0].CidrBlock" --output text; aws ec2 describe-subnets --filters "Name=vpc-id,Values=${VPC_ID}" --query "Subnets[].{Name:Tags[?Key=='Name'].Value|[0],CIDR:CidrBlock}" --output text 2) VPC와 서브넷의 이름이 정확하게 일치하며 VPC와 서브넷에 지정한 CIDR가 부여되어있는지 확인합니다. 192.168.0.0/16 192.168.1.0/24 wsc2026-skills-hub-sub-a 192.168.10.0/24 wsc2026-skills-hub-sub-b 192.168.2.0/24 wsc2026-skills-app-sub-a 192.168.20.0/24 wsc2026-skills-app-sub-b
1-2	1) IGW_ID=$(aws ec2 describe-internet-gateways --filters "Name=attachment.vpc-id,Values=${VPC_ID}" "Name=tag:Name,Values=wsc2026-skills-igw" --query "InternetGateways[0].InternetGatewayId" --output text); echo "IGW: $IGW_ID" NAT_A_ID=$(aws ec2 describe-nat-gateways --filter "Name=vpc-id,Values=${VPC_ID}" "Name=tag:Name,Values=wsc2026-skills-nat-a" "Name=state,Values=available" --query "NatGateways[0].NatGatewayId" --output text); NAT_B_ID=$(aws ec2 describe-nat-gateways --filter "Name=vpc-id,Values=${VPC_ID}" "Name=tag:Name,Values=wsc2026-skills-nat-b" "Name=state,Values=available" --query "NatGateways[0].NatGatewayId" --output text); echo "NAT-A: $NAT_A_ID"; echo "NAT-B: $NAT_B_ID" aws ec2 describe-route-tables --filters "Name=vpc-id,Values=${VPC_ID}" "Name=tag:Name,Values=*" --query "RouteTables[].{Name:Tags[?Key=='Name'].Value|[0],Route:Routes[?DestinationCidrBlock=='0.0.0.0/0']|[0]}" --output text 2) 실행 결과가 강조된 텍스트와 동일하며, hub-rtb에는 igw의 ID가, app-rtb-a에는 NAT-A의 ID가, app-rtb-b에는 NAT-B의 ID가 서로 일치하게 연결되어 있는지 확인합니다. IGW: igw-0a85cd67732857621 NAT-A: nat-071e3d4820939525d NAT-B: nat-03f60025e555f65bb wsc2026-skills-app-rtb-b ROUTE 0.0.0.0/0 nat-03f60025e555f65bb CreateRoute active wsc2026-skills-hub-rtb ROUTE 0.0.0.0/0 igw-0a85cd67732857621 CreateRoute active wsc2026-skills-app-rtb-a ROUTE 0.0.0.0/0 nat-071e3d4820939525d CreateRoute active

순번	채점 항목
2-1	1) aws dynamodb describe-table --table-name wsc2026-book-table --query "Table.[KeySchema[0].AttributeName,BillingModeSummary.BillingMode,SSEDescription.SSEType,DeletionProtectionEnabled,GlobalSecondaryIndexes[0].KeySchema[0].AttributeName]" --output text 2>/dev/null aws dynamodb describe-continuous-backups --table-name wsc2026-book-table --query "ContinuousBackupsDescription.[PointInTimeRecoveryDescription.PointInTimeRecoveryStatus,PointInTimeRecoveryDescription.RecoveryPeriodInDays,ContinuousBackupsStatus]" --output text 2>/dev/null aws dynamodb get-resource-policy --resource-arn "arn:aws:dynamodb:ap-northeast-2:${ACCOUNT_ID}:table/wsc2026-book-table" --query "Policy" --output text 2>/dev/null \| python3 -c "import sys,json;[print(f'{s.get(W"ActionW",W"W")} : {s.get(W"PrincipalW",{}).get(W"AWSW",W"W").split(W"/W")[-1]}') for s in json.loads(sys.stdin.read()).get('Statement',[])]" 2>/dev/null \|\| echo "No resource policy" check_kms "wsc2026-db-kms" "$(aws dynamodb describe-table --table-name wsc2026-book-table --query 'Table.SSEDescription.KMSMasterKeyArn' --output text 2>/dev/null)" 2) 실행 결과가 아래와 정확히 일치해야 합니다. 단, 강조된 부분의 출력 순서는 무관합니다. client_id PAY_PER_REQUEST KMS True booking_id ENABLED 35 ENABLED **dynamodb:PutItem : wsc2026-book-pod-role** **dynamodb:Query : wsc2026-book-function-role** KMS wsc2026-db-kms: PASS
3-1	1) aws ecr describe-repositories --repository-names wsc2026-book-ecr --query "repositories[0].[imageScanningConfiguration.scanOnPush,imageTagMutability,imageTagMutabilityExclusionFilters[0].filter,encryptionConfiguration.encryptionType]" --output text 2>/dev/null; aws ecr list-images --repository-name wsc2026-book-ecr --query "imageIds[].imageTag" --output text 2>/dev/null check_kms "wsc2026-ecr-kms" "$(aws ecr describe-repositories --repository-names wsc2026-book-ecr --query 'repositories[0].encryptionConfiguration.kmsKey' --output text 2>/dev/null)" 2) 실행 결과가 아래와 동일한지 확인합니다. True MUTABLE_WITH_EXCLUSION v1* KMS v1.0.0 KMS wsc2026-ecr-kms: PASS

순번	채점 항목
4-1	1) aws eks describe-cluster --name wsc2026-eks-cluster --query "cluster.[version,status,resourcesVpcConfig.endpointPublicAccess,resourcesVpcConfig.endpointPrivateAccess,logging.clusterLogging[0].enabled]" --output text 2>/dev/null CLUSTER_SG=$(aws eks describe-cluster --name wsc2026-eks-cluster --query "cluster.resourcesVpcConfig.clusterSecurityGroupId" --output text 2>/dev/null); ANYIP=$(aws ec2 describe-security-groups --group-ids "$CLUSTER_SG" --query "SecurityGroups[0].IpPermissions[?IpProtocol=='-1'].IpRanges[].CidrIp" --output text 2>/dev/null); if [-n "$ANYIP"]; then echo "SG: FAIL"; else echo "SG: PASS"; fi check_kms "wsc2026-eks-kms" "$(aws eks describe-cluster --name wsc2026-eks-cluster --query 'cluster.encryptionConfig[0].provider.keyArn' --output text 2>/dev/null)" 2) 실행 결과가 아래와 동일한지 확인합니다. 1.35 ACTIVE False True True SG: PASS wsc2026.skills.local KMS wsc2026-eks-kms: PASS
4-2	1) NG1=$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-addon-nodegroup --query "nodegroup.[nodegroupName,instanceTypes[0]]" --output text 2>/dev/null); echo "$NG1 wsc2026/node=addon" ADDON_COUNT=$(kubectl get nodes -l wsc2026/node=addon --no-headers --request-timeout=10s 2>/dev/null \| wc -l) if ["$ADDON_COUNT" -ge 1]; then echo "Addon Nodes: PASS ($ADDON_COUNT)"; else echo "Addon Nodes: FAIL"; fi NG2=$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-workload-ng --query "nodegroup.[nodegroupName,instanceTypes[0]]" --output text 2>/dev/null); echo "$NG2 wsc2026/node=application" WORK_COUNT=$(kubectl get nodes -l wsc2026/node=application --no-headers --request-timeout=10s 2>/dev/null | wc -l) if ["$WORK_COUNT" -ge 1]; then echo "Workload Nodes: PASS ($WORK_COUNT)"; else echo "Workload Nodes: FAIL"; fi 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-addon-nodegroup t3.medium wsc2026/node=addon Addon Nodes: PASS (2) wsc2026-workload-ng t3.medium wsc2026/node=application Workload Nodes: PASS (2)

순번	채점 항목
4-3	1) CR=$(aws eks describe-cluster --name wsc2026-eks-cluster --query "cluster.roleArn" --output text 2>/dev/null | awk -F/ '{print $NF}'); AR=$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-addon-nodegroup --query "nodegroup.nodeRole" --output text 2>/dev/null | awk -F/ '{print $NF}'); WR=$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-workload-ng --query "nodegroup.nodeRole" --output text 2>/dev/null | awk -F/ '{print $NF}') for PAIR in "Cluster Role:$CR" "Addon Node Role:$AR" "Workload Node Role:$WR"; do LABEL="${PAIR%%:*}"; ROLE="${PAIR#*:}"; ADMIN=$(aws iam list-attached-role-policies --role-name "$ROLE" --query "AttachedPolicies[?PolicyName=='AdministratorAccess'].PolicyName" --output text 2>/dev/null); if [-n "$ADMIN"]; then echo "$LABEL: FAIL"; else echo "$LABEL: PASS"; fi; done 2) 실행 결과가 아래와 동일한지 확인합니다. Cluster Role: PASS Addon Node Role: PASS Workload Node Role: PASS
5-1	1) kubectl get deploy,svc,ingress,pdb -n wsc2026 --no-headers 2>/dev/null | awk ' /^deployment/ { split($2,a,"/"); print "Deployment: " (a[1]==a[2] && a[1]=="2" ? "PASS" : "FAIL") } /^service/ { print "Service: PASS" } /^ingress/ { print "Ingress: " ($4 ~ /W.elbW.amazonawsW.com/ ? "PASS ("$4")" : "FAIL") } /^poddisrupt/ { print "PDB: " ($2=="1" ? "PASS" : "FAIL") } ' 2) 실행결과가 아래와 동일한지 확인하며, Ingress의 ALB DNS는 wsc2026-app-alb가 포함되었는지 확인합니다. Deployment: PASS Service: PASS Ingress: PASS (wsc2026-app-alb-829903010.ap-northeast-2.elb.amazonaws.com) PDB: PASS
5-2	1) kubectl get deploy wsc2026-book-deploy -n wsc2026 -o jsonpath='replicas:{.spec.replicas} node:{.spec.template.spec.nodeSelector.wsc2026/node} topo:{.spec.template.spec.topologySpreadConstraints[0].topologyKey} cpu:{.spec.template.spec.containers[0].resources.requests.cpu} mem:{.spec.template.spec.containers[0].resources.requests.memory}' 2>/dev/null 2) 실행 결과가 아래와 동일한지 확인합니다. replicas:2 node:application topo:topology.kubernetes.io/zone cpu:250m mem:512Mi

순번	채점 항목
5-3	1) kubectl get deploy wsc2026-book-deploy -n wsc2026 -o jsonpath='startup:{.spec.template.spec.containers[0].startupProbe.httpGet.path}:{.spec.template.spec.containers[0].startupProbe.httpGet.port} readiness:{.spec.template.spec.containers[0].readinessProbe.httpGet.path}:{.spec.template.spec.containers[0].readinessProbe.httpGet.port} liveness:{.spec.template.spec.containers[0].livenessProbe.httpGet.path}:{.spec.template.spec.containers[0].livenessProbe.httpGet.port}' 2>/dev/null; echo; kubectl get configmap book-config -n wsc2026 -o jsonpath='{.data}' 2>/dev/null; echo 2) 실행 결과가 아래와 동일한지 확인합니다. startup:/health:8080 readiness:/health:8080 liveness:/health:8080 {"AWS_REGION":"ap-northeast-2","TABLE_NAME":"wsc2026-book-table"}
5-4	1) WORKLOAD_NODES=$(kubectl get nodes -l wsc2026/node=application --no-headers -o custom-columns='NAME:.metadata.name' 2>/dev/null) kubectl get pods -n wsc2026 -o custom-columns='NAME:.metadata.name,STATUS:.status.phase,NODE:.spec.nodeName' --no-headers 2>/dev/null \| while read NAME STATUS NODE; do echo "$NAME: $(echo "$WORKLOAD_NODES" | grep -q "^$NODE$" && echo PASS || echo FAIL)" done 2) 실행 결과에서 wsc2026-book-deploy가 포함된 텍스트가 2개 출력되며, 모두 PASS인지 확인합니다. wsc2026-book-deploy-7b88597d86-d6l72: PASS wsc2026-book-deploy-7b88597d86-z8nkt: PASS
5-5	1) PI_SA=$(aws eks list-pod-identity-associations --cluster-name wsc2026-eks-cluster --namespace wsc2026 --query 'associations[0].serviceAccount' --output text 2>/dev/null) PI_ROLE=$(aws eks list-pod-identity-associations --cluster-name wsc2026-eks-cluster --namespace wsc2026 --query 'associations[0].associationId' --output text 2>/dev/null) ["$PI_SA" = "wsc2026-book-sa"] && echo "Pod Identity SA: PASS ($PI_SA)" || echo "Pod Identity SA: FAIL ($PI_SA)" ROLE_NAME=$(aws eks describe-pod-identity-association --cluster-name wsc2026-eks-cluster --association-id "$PI_ROLE" --query 'association.roleArn' --output text 2>/dev/null|awk -F/ '{print $NF}') MANAGED_ACTIONS=$(aws iam list-attached-role-policies --role-name "$ROLE_NAME" --query 'AttachedPolicies[].PolicyArn' --output text 2>/dev/null\|tr '₩t' '₩n'\|while read arn;do VERSION=$(aws iam get-policy --policy-arn "$arn" --query 'Policy.DefaultVersionId' --output text 2>/dev/null);aws iam get-policy-version --policy-arn "$arn" --version-id "$VERSION" --query 'PolicyVersion.Document.Statement[].Action' --output text 2>/dev/null;done) echo "$MANAGED_ACTIONS"|grep -q dynamodb:PutItem && ! echo "$MANAGED_ACTIONS"\|grep -qE '^₩*$' && echo "Pod Identity Role: PASS" || echo "Pod Identity Role: FAIL" 2) 실행 결과가 아래와 동일한지 확인합니다. Pod Identity SA: PASS (wsc2026-book-sa) Pod Identity Role: PASS

순번	채점 항목

6-1	1) echo "$BUCKET"; aws s3api get-public-access-block --bucket "$BUCKET" --query "PublicAccessBlockConfiguration.[BlockPublicAcls,BlockPublicPolicy,IgnorePublicAcls,RestrictPublicBuckets]" --output text 2>/dev/null; aws s3api get-bucket-encryption --bucket "$BUCKET" --query "ServerSideEncryptionConfiguration.Rules[0].{SSE:ApplyServerSideEncryptionByDefault.SSEAlgorithm,BucketKey:BucketKeyEnabled}" --output text 2>/dev/null; aws s3api list-objects --bucket "$BUCKET" --prefix "static/" --query "Contents[?Size>W`0W`].Key" --output text 2>/dev/null
	check_kms "wsc2026-bucket-kms" "$(aws s3api get-bucket-encryption --bucket "$BUCKET" --query 'ServerSideEncryptionConfiguration.Rules[0].ApplyServerSideEncryptionByDefault.KMSMasterKeyID' --output text 2>/dev/null)"
	BUCKET_KMS_ARN=$(aws kms describe-key --key-id "alias/wsc2026-bucket-kms" --query "KeyMetadata.Arn" --output text 2>/dev/null)
	echo "S3 Object KMS Check:"; for OBJ in $(aws s3api list-objects --bucket "$BUCKET" --prefix "static/" --query "Contents[].Key" --output text 2>/dev/null); do
	  KEY_ID=$(aws s3api head-object --bucket "$BUCKET" --key "$OBJ" --query "SSEKMSKeyId" --output text 2>/dev/null)
	  if ["$KEY_ID" = "$BUCKET_KMS_ARN"]; then echo " $OBJ: PASS"; else echo " $OBJ: FAIL ($KEY_ID)"; fi
	done
	2) 실행 결과가 아래와 동일한지 확인합니다.
	wsc2026-static-<임의 영문 4자리>-<선수 비번호>-bucket
	True True True True
	True aws:kms
	static/index.html static/main.jpeg
	KMS wsc2026-bucket-kms: PASS
	S3 Object KMS Check:
	~~static/: PASS~~
	  static/index.html: PASS
	  static/main.jpeg: PASS

7-1	1) aws lambda get-function --function-name wsc2026-book-get-function --query "Configuration.{Runtime:Runtime,Env:Environment.Variables.TABLE_NAME}" --output text 2>/dev/null
	check_kms "wsc2026-function-kms" "$(aws lambda get-function --function-name wsc2026-book-get-function --query 'Configuration.KMSKeyArn' --output text 2>/dev/null)"
	2) 실행 결과가 아래와 동일한지 확인하며, TABLE_NAME은 평문이 아닌 KMS로 암호화된 텍스트가 출력되는지 확인합니다.
	{
	  "Name": "wsc2026-book-get-function",
	  "Runtime": "python3.12"
	}
	{
	  "TABLE_NAME": "AQICAHjHYzx8HkYlOX8fnCJbPqPanGMX7gxnAX57CFiSG1wT1wEpFZotnxm3...."
	}
	KMS wsc2026-function-kms: PASS

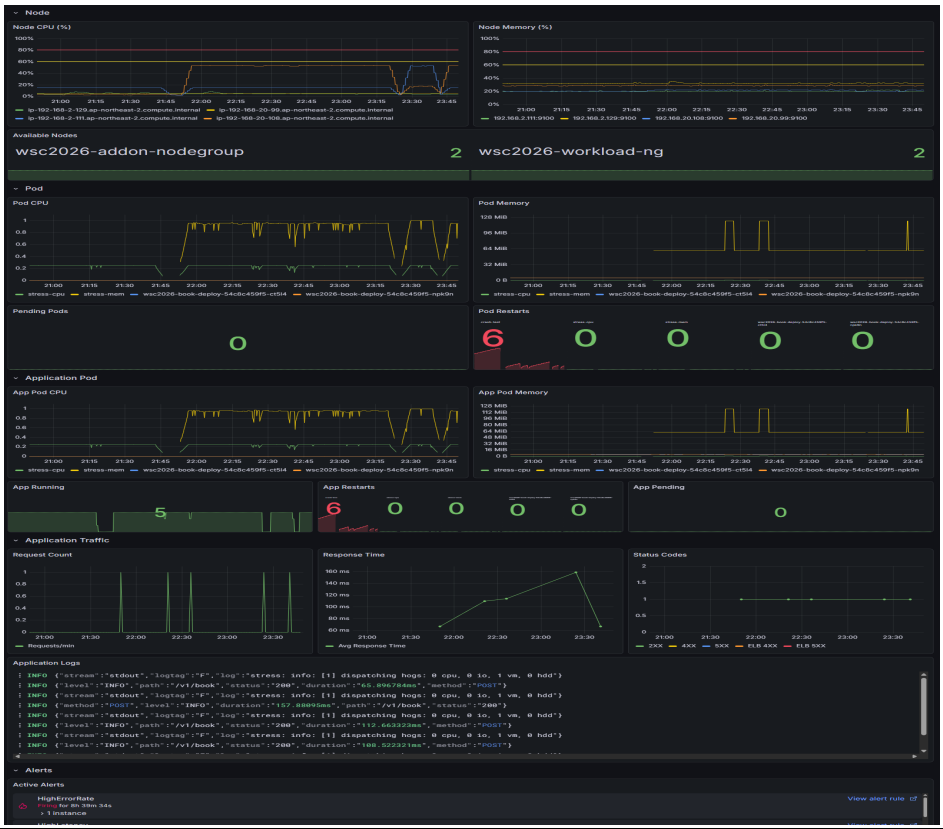
순번	채점 항목
7-2	1) LAMBDA_ROLE=$(aws lambda get-function --function-name wsc2026-book-get-function --query "Configuration.Role" --output text 2>/dev/null | awk -F/ '{print $NF}'); echo "$LAMBDA_ROLE"; POLICIES=$(aws iam list-attached-role-policies --role-name "$LAMBDA_ROLE" --query "AttachedPolicies[].PolicyName" --output text 2>/dev/null); echo "$POLICIES"; if echo "$POLICIES" | grep -q "AdministratorAccess"; then echo "Role: FAIL (Admin)"; else echo "Role: PASS"; fi POLICY_ARN=$(aws iam list-attached-role-policies --role-name "$LAMBDA_ROLE" --query "AttachedPolicies[?PolicyName!='AWSLambdaBasicExecutionRole'].PolicyArn|[0]" --output text 2>/dev/null); ACTIONS=$(aws iam get-policy-version --policy-arn "$POLICY_ARN" --version-id $(aws iam get-policy --policy-arn "$POLICY_ARN" --query "Policy.DefaultVersionId" --output text 2>/dev/null) --query "PolicyVersion.Document.Statement[].Action" --output text 2>/dev/null) if echo "$ACTIONS" \| grep -q "dynamodb:Query" && ! echo "$ACTIONS" | grep -q 'W*'; then echo "Policy: PASS"; else echo "Policy: FAIL"; fi 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-book-function-role wsc2026-book-function-policy Role: PASS Policy: PASS
8-1	1) aws elbv2 describe-load-balancers --names wsc2026-app-alb --query "LoadBalancers[0].Scheme" --output text 2>/dev/null; ALB_SGS=$(aws elbv2 describe-load-balancers --names wsc2026-app-alb --query "LoadBalancers[0].SecurityGroups[]" --output text 2>/dev/null); aws ec2 describe-security-groups --group-ids $ALB_SGS --query "SecurityGroups[].GroupName" --output text 2>/dev/null ALB_DNS=$(aws elbv2 describe-load-balancers --names wsc2026-app-alb --query "LoadBalancers[0].DNSName" --output text 2>/dev/null); ALB_CODE=$(curl -s -o /dev/null -w "%{http_code}" --max-time 5 "http://${ALB_DNS}/" 2>/dev/null); if ["$ALB_CODE" = "000"]; then echo "ALB direct: BLOCKED"; else echo "ALB direct: FAIL ($ALB_CODE)"; fi 2) 실행 결과가 아래와 동일한지 확인합니다. internet-facing wsc2026-app-alb-sg ALB direct: BLOCKED
9-1	1) CF_ARN=$(aws resourcegroupstaggingapi get-resources --tag-filters Key=Name,Values=wsc2026-cdn --resource-type-filters cloudfront --region us-east-1 --query "ResourceTagMappingList[0].ResourceARN" --output text 2>/dev/null); CF_ID=$(echo "$CF_ARN" | awk -F/ '{print $NF}'); CF_DOMAIN=$(aws cloudfront get-distribution --id "$CF_ID" --region us-east-1 --query "Distribution.DomainName" --output text 2>/dev/null); curl -s -o /dev/null -w "${CF_DOMAIN} : %{http_code}" "https://${CF_DOMAIN}/" 2>/dev/null; echo 2) 실행 결과에서 Cloudfront 도메인과 200이 출력되는지 확인합니다. d1ain3jwa2akkh.cloudfront.net : 200

순번	채점 항목
9-2	1) DISABLED="4135ea2d-6df8-44a3-9df3-4b5a84be39ad"; OPTIMIZED="658327ea-f89d-4fab-a63d-7e88639e58f6"; DEFAULT_CP=$(aws cloudfront get-distribution --id "$CF_ID" --region us-east-1 --query "Distribution.DistributionConfig.DefaultCacheBehavior.CachePolicyId" --output text 2>/dev/null); if ["$DEFAULT_CP" = "$OPTIMIZED"]; then echo "S3: CachingOptimized"; else echo "S3: FAIL"; fi for B in $(aws cloudfront get-distribution --id "$CF_ID" --region us-east-1 --query "Distribution.DistributionConfig.CacheBehaviors.Items[].PathPattern" --output text 2>/dev/null); do CP=$(aws cloudfront get-distribution --id "$CF_ID" --region us-east-1 --query "Distribution.DistributionConfig.CacheBehaviors.Items[?PathPattern=='${B}'].CachePolicyId|[0]" --output text 2>/dev/null); if ["$CP" = "$DISABLED"]; then echo "ALB/Lambda: CachingDisabled"; break; fi; done 2) 실행 결과가 아래와 동일한지 확인합니다. S3: CachingOptimized ALB/Lambda: CachingDisabled
9-3	1) BOOKING_ID=$(curl -s -X POST "https://${CF_DOMAIN}/booking" -H "Content-Type: application/json" -d '{"client_id":"MARK001","username":"Marker","email":"mark@test.com","concert_name":"TestConcert"}' 2>/dev/null | python3 -c "import sys,json;print(json.load(sys.stdin).get('booking_id',''))" 2>/dev/null); echo "POST booking_id: $BOOKING_ID" curl -s "https://${CF_DOMAIN}/v1/book?booking_id=${BOOKING_ID}" 2>/dev/null; echo 2) 필드 순서(client_id, username, email, concert_name, created_at)가 차례대로 일치하는지, 그리고 created_at 시간이 스크립트 실행 시간 기준 1분 이내인지 확인합니다. POST booking_id: NWOEMIU2 {"client_id": "MARK001", "username": "Marker", "email": "mark@test.com", "concert_name": "TestConcert", "created_at": "2026-05-15 00:13:05 KST"}
10-1	1) WAF_NAME=$(aws wafv2 list-web-acls --scope CLOUDFRONT --region us-east-1 --query "WebACLs[?contains(Name, 'wsc2026')].Name" --output text 2>/dev/null) echo "WAF Name: $WAF_NAME" SQLi=$(curl -s -o /dev/null -w '%{http_code}' --max-time 5 "https://${CF_DOMAIN}/v1/book?booking_id=1'%20OR%201=1--"); echo "SQLi: $SQLi" XSS=$(curl -s -o /dev/null -w '%{http_code}' --max-time 5 "https://${CF_DOMAIN}/v1/book?booking_id=<script>alert(1)</script>"); echo "XSS: $XSS" (for i in $(seq 1 15); do for j in $(seq 1 25); do curl -s -o /dev/null --max-time 3 "https://${CF_DOMAIN}/" & done; sleep 0.4; done; wait) >/dev/null 2>&1; WAF_ID=$(aws wafv2 list-web-acls --scope CLOUDFRONT --region us-east-1 --query "WebACLs[?contains(Name, 'wsc2026')].Id" --output text 2>/dev/null); RATE_LIMIT=$(aws wafv2 get-web-acl --scope CLOUDFRONT --region us-east-1 --name "$WAF_NAME" --id "$WAF_ID" --query "WebACL.Rules[?Statement.RateBasedStatement].Statement.RateBasedStatement.Limit" --output text 2>/dev/null); if [-n "$RATE_LIMIT"] && ["$RATE_LIMIT" -le 200] 2>/dev/null; then echo "Rate: PASS (403)"; else echo "Rate: FAIL (-)"; fi 2) 실행 결과가 아래와 동일한지 확인합니다. WAF Name: wsc2026-waf SQLi: 403 XSS: 403 Rate: PASS (403)

순번	채점 항목
11-1	1) ~~SVC_IP=$(kubectl get svc -n wsc2026 -o jsonpath='{.items[0].spec.clusterIP}' 2>/dev/null); kubectl run not-ready --image=busybox --restart=Always -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"},"containers":[{"name":"not-ready","image":"busybox","readinessProbe":{"httpGet":{"path":"/health","port":80},"periodSeconds":3},"command":["sh","-c","sleep 3600"]}]}}' &>/dev/null; kubectl run error-gen --image=curlimages/curl --restart=Never -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"}}}' -- sh -c "while true; do curl -s -o /dev/null http://'"$SVC_IP"'/nonexist; sleep 0.1; done" &>/dev/null; kubectl run latency-gen --image=curlimages/curl --restart=Never -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"}}}' -- sh -c "while true; do curl -s -o /dev/null http://'"$SVC_IP"'/delay?ms=5000; sleep 0.2; done" &>/dev/null~~ ~~kubectl run crash-test --image=busybox --restart=Always -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"}}}' -- sh -c 'exit 1' &>/dev/null; kubectl run stress-cpu --image=busybox --restart=Never -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"},"containers":[{"name":"stress-cpu","image":"busybox","resources":{"requests":{"cpu":"250m"},"limits":{"cpu":"250m"}},"command":["sh","-c","while true; do :; done"]}]}}' &>/dev/null; kubectl run stress-mem --image=polinux/stress --restart=Never -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"},"containers":[{"name":"stress-mem","image":"polinux/stress","resources":{"requests":{"memory":"64Mi"},"limits":{"memory":"64Mi"}},"command":["stress","--vm","1","--vm-bytes","60M","--vm-keep","-t","3600"]}]}}' &>/dev/null~~ ~~sleep 180~~ ~~GRAFANA_LB=$(kubectl get svc -n observability -o jsonpath='{range .items[?(@.spec.type=="LoadBalancer")]}{.status.loadBalancer.ingress[0].hostname}{end}' 2>/dev/null)~~ for p in fluent-bit prometheus grafana; do kubectl get pods -n observability --no-headers --request-timeout=10s 2>/dev/null | grep -c "$p.*Running" \| xargs -I{} echo "$p: {}"; done 2) 실행 결과가 1 이상인지 확인합니다. fluent-bit: 4 prometheus: 7 grafana: 1

11-2	1) echo "Datasources:"; curl -s -u admin:'Skills$#$@!' "http://${GRAFANA_LB}/api/datasources" 2>/dev/null | python3 -c "import sys,json;[print(f' {d[W"nameW"]} ({d[W"typeW"]})') for d in json.load(sys.stdin)]" 2>/dev/null || echo " Grafana unreachable"
	echo "Dashboards:"; curl -s -u admin:'Skills$#$@!' "http://${GRAFANA_LB}/api/search?query=wsc2026" 2>/dev/null | python3 -c "import sys,json;[print(f' {d[W"titleW"]}') for d in json.load(sys.stdin)]" 2>/dev/null || echo " Not found"
	2) 실행 결과가 아래와 동일한지 확인합니다.
	Datasources:
	  alertmanager (alertmanager)
	  cloudwatch (cloudwatch)
	  prometheus (prometheus)
	Dashboards:
	  wsc2026-grafana-dashboard

순번	채점 항목
11-3	1) 아래 명령을 수동으로 입력합니다. SVC_IP=$(kubectl get svc -n wsc2026 -o jsonpath='{.items[0].spec.clusterIP}' 2>/dev/null); kubectl run not-ready --image=busybox --restart=Always -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"},"containers":[{"name":"not-ready","image":"busybox","readinessProbe":{"httpGet":{"path":"/health","port":80},"periodSeconds":3},"command":["sh","-c","sleep 3600"]}]}}' &>/dev/null; kubectl run error-gen --image=curlimages/curl --restart=Never -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"}}}' -- sh -c "while true; do curl -s -o /dev/null http://'"$SVC_IP"'/nonexist; sleep 0.1; done" &>/dev/null; kubectl run latency-gen --image=curlimages/curl --restart=Never -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"}}}' -- sh -c "while true; do curl -s -o /dev/null http://'"$SVC_IP"'/delay?ms=5000; sleep 0.2; done" &>/dev/null kubectl run crash-test --image=busybox --restart=Always -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"}}}' -- sh -c 'exit 1' &>/dev/null; kubectl run stress-cpu --image=busybox --restart=Never -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"},"containers":[{"name":"stress-cpu","image":"busybox","resources":{"requests":{"cpu":"250m"},"limits":{"cpu":"250m"}},"command":["sh","-c","while true; do :; done"]}]}}' &>/dev/null; kubectl run stress-mem --image=polinux/stress --restart=Never -n wsc2026 --overrides='{"spec":{"tolerations":[{"operator":"Exists"}],"nodeSelector":{"wsc2026/node":"application"},"containers":[{"name":"stress-mem","image":"polinux/stress","resources":{"requests":{"memory":"64Mi"},"limits":{"memory":"64Mi"}},"command":["stress","--vm","1","--vm-bytes","60M","--vm-keep","-t","3600"]}]}}' &>/dev/null sleep 180 GRAFANA_LB=$(kubectl get svc -n observability -o jsonpath='{range .items[?(@.spec.type=="LoadBalancer")]}{.status.loadBalancer.ingress[0].hostname}{end}' 2>/dev/null) ... 다음 페이지 이어서

순번	채점 항목
11-3	2) 출력된 Grafana URL을 통해 admin/Skills$#$@!에 로그인 후 wsc2026-grafana-dashboard에서 ~~메트릭과 로그 형식이 채점지 사진과 일치하는지 확인하며~~, 모든 메트릭 **및 로그는** 빈값이 없어야합니다. **다음 구성 요소가 대쉬보드에 포함되어 있는지 확인합니다.** **- Node CPU (%) - 노드별 CPU 사용률** **- Node Memory (%) - 노드별 Memory 사용률** **- Available Nodes - 각 노드그룹별 노드 수** **- Pod CPU - 파드별 CPU 사용률** **- Pod Memory - 파드별 Memory 사용률** **- Pending Pods - Pending 상태의 Pod 개수** **- Pod Restarts - 리스타트 Pod 개수** **- App Pod CPU - 어플리케이션 파드별 CPU 사용률** **- App Pod Memory - 어플리케이션 파드별 Memory 사용률** **- App Running - 어플리케이션 파드 Running 상태의 수** **- App Restarts - 어플리케이션 파드 리스타트 수** **- App Pending - 어플리케이션 파드 Pending 상태의 수** **- Request Count - 전체 어플리케이션 요청 수** **- Response Time - 전체 어플리케이션 평균 응답 시간** **- Status Code - 어플리케이션 응답 코드별 응답 수** **- Application Logs - 어플리케이션 로그** - Application Logs 패널 로그 형식 예시 (/v1/book을 제외한 로그가 있을 경우 오답처리): INFO {"level":"INFO","path":"/v1/book","status":"200","duration":"112.663323ms","method":"POST"} **(참고 사진)** 

순번	채점 항목
11-4	1) Active Alerts 패널에 Edit을 선택해 나온 결과와 채점지 사진과 일치하는지 확인합니다. **(* HighLatency Alert 구성요소는 채점과 무관합니다.)**